



DFDOF

Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	ft-df006-2017-ios-2
Operator:	Floris
Analysis Started:	2026-06-19 11:15:06
Analysis Completed:	2026-06-19 11:22:53
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\test_phantom_4\df006\2017\ios
Output Directory:	C:\Users\Floris\Documents\dji_drones\test_phantom_4\df006\2017\dfdof_output\ft-df006-2017-ios-2
Evidence Sources:	3
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
df006_external.001	drone_sd	physical	14.9 GB	2026-06-19 11:16:01
DF006_flight.zip	controller_ios	logical	947.3 MB	2026-06-19 11:16:03
df006_internal.001	drone_flight_storage	physical	7.4 GB	2026-06-19 11:16:26

Evidence Source Path

C:\Users\Floris\Documents\ddji_drones\test_phantom_4\df006\2017\ios

Cryptographic Hashes

df006_external.001		drone_sd	physical
SHA-256:	004071f5ee6d64cb85a6632b95720a03f0b662a2b1c565a1d8fa0b6249df3ab7		
SHA-1:	e1e054bec2474156b046e0e68e4ed31465176ef1		
DF006_flight.zip		controller_ios	logical
SHA-256:	8a25a5dc0debabab7cd893e44bd824958cf7ef01cd002c1459d529c4c55a9526		
SHA-1:	b9dc6bde2a851a24e16b9fc3ee7183abc42d9693		
df006_internal.001		drone_flight_storage	physical
SHA-256:	53cb72f413544274315a6d468969802a4da401a4a2f342b78ea5cec4c07e0366		
SHA-1:	bd6eb1c7396b606e494fd144eded4cfcf7d5e853		

Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p3 - controller android]: source evidence not found
3. [p3 - controller ios - drone logs]: no artefacts found
4. [p4 - controller android]: source evidence not found
5. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY000.csv
6. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY001.csv
7. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY004.csv
8. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY009.csv
9. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY010.csv

Tool Execution Status

Tool	Invocations	Successes	Status
datcon	11	11	[OK]
exiftool	43	43	[OK]
fls	2	2	[OK]
icat	40	40	[OK]
mmls	2	2	[OK]
parser_ios	1	1	[OK]
txtlogtocsvtool	1	1	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	3 / 4
Artefact Categories With Data	7 / 7
Flights With Primary Correlation	0 / 7
Tools Succeeded	7 / 7

3. Device and Account Identification

Field	Value	Confidence	Sources
Installed DJI App(s)	com.dji.go com.dji.pilot	[K] Controller-only	1
Controller Device	iPad mini 4	[K] Controller-only	1
Drone Serial Number	0BMLE5501503RX	[K] Controller-only	1
Drone Model	P4P Spark	[!] Inconsistent	7
Controller Device	iPad5,1__weibosdk__003143000__iphone__os 10.3.2	[K] Controller-only	2
DJI Application Version	3.1.12 4.1.3	[!] Inconsistent	3
Account E-Mail	droneforensics@vtoinc.com	[K] Controller-only	2
Product Type	2 26	[!] Inconsistent	2
Last Country Code	US	[K] Controller-only	1
Last App Launch	2017-06-28T19:55:11Z 2017-06-29T16:40:17Z	[!] Inconsistent	2
Last Flight Date	1970-01-01T00:00:00Z	[K] Controller-only	2

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: acaa4c8dc5826e59f451bbbd7f40c9d2c4003b8e74e8709cc78835cefde155a5
Drone Log Name: FLY002.csv
Start: 2017-06-28T17:43:55.558000+00:00
End: 2017-06-28T17:54:29.423000+00:00
Duration: 633.9 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

1. Manual
2. Atti
3. GPS_Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Reached peak height	-0.0 m
-	drone_flight_storage:dl	Motor turned off	off
2017-06-28 17:50:23	drone_flight_storage:dl	Log started	P4P
2017-06-28 17:54:29	drone_flight_storage:dl	Log ended	

Correlated Artefacts

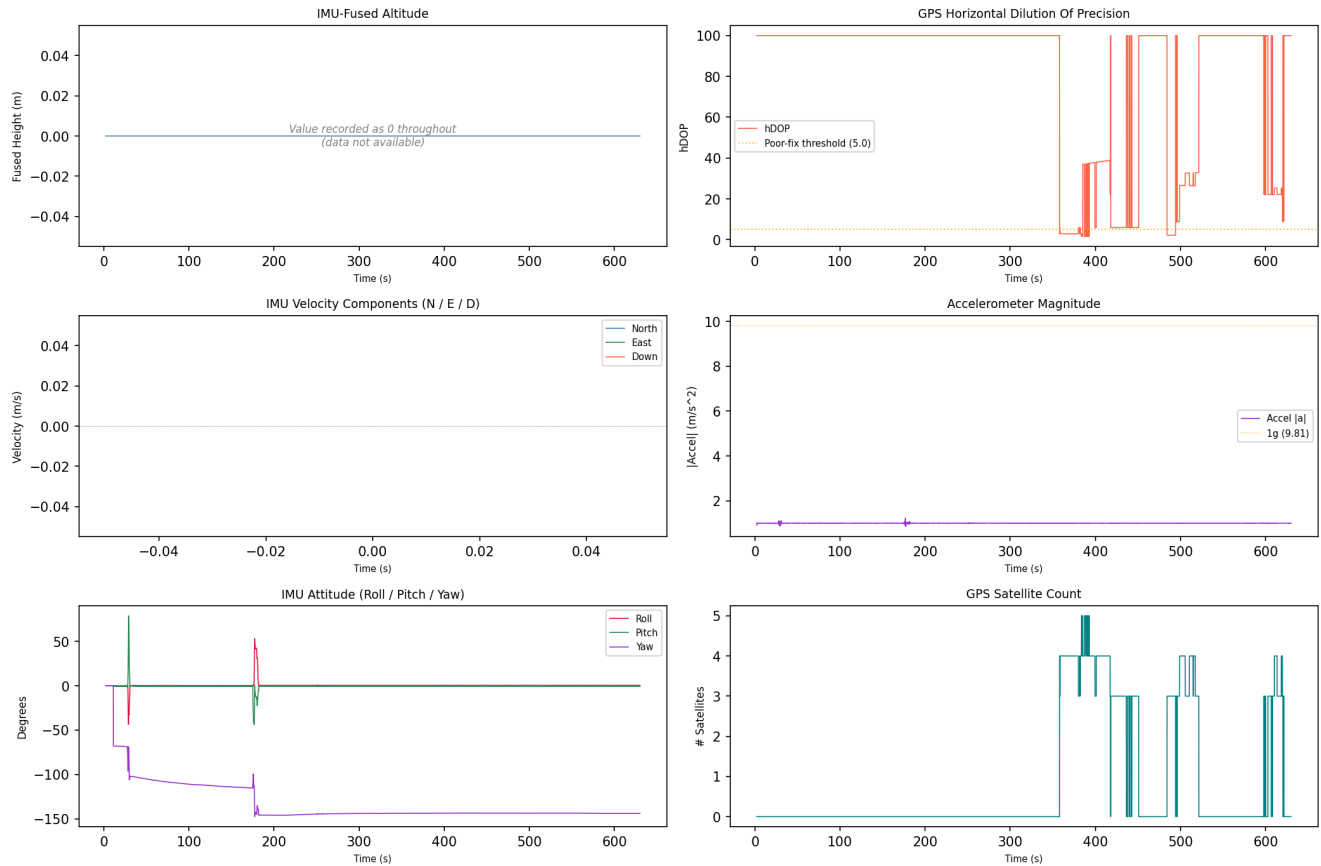
Possibly correlated (indirect indicators):

drone_sd:images	GPS within flight bounding box (lat 39.9612, lon 106.2165)
p5:ca120108a8b18e5deb186db4ff0a0783f13be9ec340a609529c550cd565b3900	
drone_sd:images	GPS within flight bounding box (lat 39.9574, lon 106.2177)
p5:156179a4c036f4ff09313fa145042228cdcee0792ea1687dd2bfb714a472a7ea	
drone_sd:images	GPS within flight bounding box (lat 39.9574, lon 106.2177)
p5:b6ccdc3903027f542d8fd0379c4f1001f994c638a5d5030195bbc8a7a14468fc	
drone_sd:images	GPS within flight bounding box (lat 39.9587, lon 106.2203)
p5:ba6f37f102e847211489b5371fdb817329d855f566a132e310763588a20b20e	
drone_sd:images	GPS within flight bounding box (lat 39.9607, lon 106.2209)
p5:01d51be119db0b02d8d7b7e31a01b115411c26dfe177eb66b5dfc3de6d60868c	
drone_sd:images	GPS within flight bounding box (lat 39.9607, lon 106.2209)
p5:aa97f908554705972fbb0362d863056286f863d4c5b17bba34190f4440f90f8b	
drone_sd:images	GPS within flight bounding box (lat 39.9637, lon 106.2207)
p5:079838ccc4ed59bd93bc6aaa74a337a092063749764e9ee9326df3b3f769107d	
drone_sd:images	GPS within flight bounding box (lat 39.9637, lon 106.2207)
p5:e00139d2a43b8ae05712f06bc14e6a0f82e64869b9c182f61c61afb888109422	
drone_sd:images	GPS within flight bounding box (lat 39.9637, lon 106.2207)
p5:0182321ece95430b6c9dbf7e27a49956dc9761c24f5cc9b03be0a9c47ad25c6d	
drone_sd:images	GPS within flight bounding box (lat 39.9638, lon 106.2208)

p5:74c0691b437ec9e3911cbdf5de1b6654c4f80a36e540b128dbb02f993df566e	
drone_sd:images	GPS within flight bounding box (lat 39.9638, lon 106.2208)
p5:f3ebdeef41ffd3b5ccf62ec7468f6f9b7d86a541bda848ecb4dd9c162f4e49e5	
drone_sd:images	GPS within flight bounding box (lat 39.9638, lon 106.2208)
p5:54922dafa9933d3fe0fb2ae7f657b3fb1d41c9516334e19e6a1b486fd0190993	
drone_sd:images	GPS within flight bounding box (lat 39.9637, lon 106.2207)
p5:fbcb22eb8455bd1f350c4a9fc9df97770072fcc727a354d2434dc1550b0b5b3bd	
drone_sd:images	GPS within flight bounding box (lat 39.9637, lon 106.2207)
p5:8d90018ed19e66acde8b90016826607c5416c0e38414b5e567e6b19d7815aa57	
drone_sd:images	GPS within flight bounding box (lat 39.9637, lon 106.2207)
p5:9041aae2121b0393c0d4c0ddf3e2f15b4f0df5249b4179431dd22c46046b416b	
drone_sd:images	GPS within flight bounding box (lat 39.9637, lon 106.2207)
p5:4169ef410cc02adc6c5eb68509a75b3da17608efc19b0ab98e77813c9cd648a0	
drone_sd:images	GPS within flight bounding box (lat 39.9637, lon 106.2207)
p5:beb7b3173644b5a8245f1ae715eda90da3639da617a4922b5afd35416f36fef8	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:b7e321ef952c4d417123970a2208663fc71b72b0b2c52068bd5bbc8cf3a511d8	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:8c0130901c10db776bb66aba8e3419a628d21c62bb208fbdbbfd02663ee668d6	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:df988a711915ffb87a43de6877693c062dbf40d2549782d7291939fb2ad44d48	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:5833218757f0835eb44ffafe533672ec0259301d097d018a6502befc59bee2f3	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:5ea89532ab3469120d74d3032d2564e63ad8b6ee42b02ce221c37d7072b256e6	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:e1d425f6213ecc84f6a68c13a8261d21acfd4d29bda3f74619d77973de0f7cff	
drone_sd:videos	GPS within flight bounding box (lat 39.9584, lon -106.2174)
p5:a796e8517cf21920e4667a415ee359f14da3f7f07a5fdda74d45b5f76c738514	
drone_sd:videos	GPS within flight bounding box (lat 39.9584, lon -106.2174)
p5:96b4e551824ffaf16a34254812778a25da71b41d93af663fe14647c64d0fc656	
drone_sd:videos	GPS within flight bounding box (lat 39.9608, lon -106.2201)
p5:b394e548cdf1e89eb132beb1f1559f69de9b2e6799fd6ad7fc70d571c36c2bff	
drone_sd:videos	GPS within flight bounding box (lat 39.9608, lon -106.2201)
p5:564e6bd58d751430e1a495a44803838717aebd0402e1da991043387d97207043	
drone_sd:videos	GPS within flight bounding box (lat 39.9607, lon -106.2209)
p5:2a11dd32eb3b2da78f99ce368820648d54b3512d61de05f5bbc1c917546f7bee	
drone_sd:videos	GPS within flight bounding box (lat 39.9607, lon -106.2209)
p5:2df4a3a5569fb636b7b982eed5386d52221e18c14cbd36b6dca1a70f8fbdeb2f	

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_01



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 02

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 663b1abdf3b629314072fdea1bc2fe59525f7ca84fa8831297f90ca3bcebbab
Drone Log Name: FLY003.csv
Start: 2017-06-28T17:55:22.762000+00:00
End: 2017-06-28T17:59:12.322000+00:00
Duration: 229.6 s

Flight Metrics

Peak Height: -
Distance Travelled: -
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

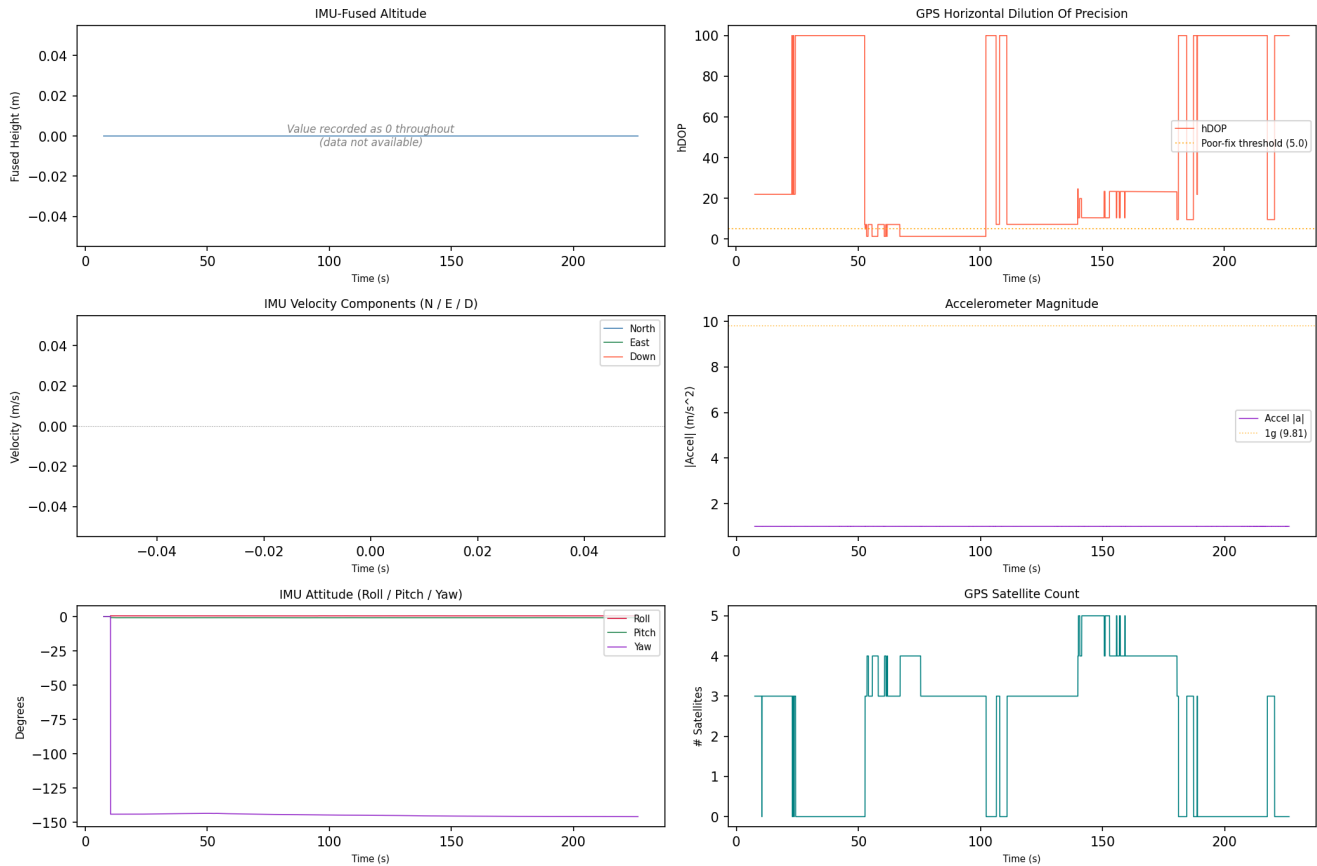
1. Atti
2. GPS_Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Reached peak height	-0.0 m
-	drone_flight_storage:dl	Motor turned off	off
2017-06-28 17:56:29	drone_flight_storage:dl	Log started	P4P
2017-06-28 17:59:12	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_02



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight track could not be plotted; view the CSV and KML files manually.

4. Flight Analysis - Flight 03

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 5a17c8e3c017d7e974ddaabc19be4e5fb6b30c45d240c0ea3ab587393ed8df5b
Drone Log Name: FLY005.csv
Start: 2017-06-29T17:01:02.991000+00:00
End: 2017-06-29T17:06:44.439000+00:00
Duration: 341.4 s

Flight Metrics

Peak Height: 0.303917021407829 m
Distance Travelled: 6.8991329738849965 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AssitedTakeoff

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2017-06-29 17:01:40	drone_flight_storage:dl	Log started	P4P
2017-06-29 17:02:25	drone_flight_storage:dl	Motor turned on	on
2017-06-29 17:02:30	drone_flight_storage:dl	Motor turned off	off
2017-06-29 17:02:33	drone_flight_storage:dl	Motor turned on	on
2017-06-29 17:02:36	drone_flight_storage:dl	Motor turned off	off
2017-06-29 17:03:21	drone_flight_storage:dl	Reached peak height	0.3 m
2017-06-29 17:06:45	drone_flight_storage:dl	Log ended	

Correlated Artefacts

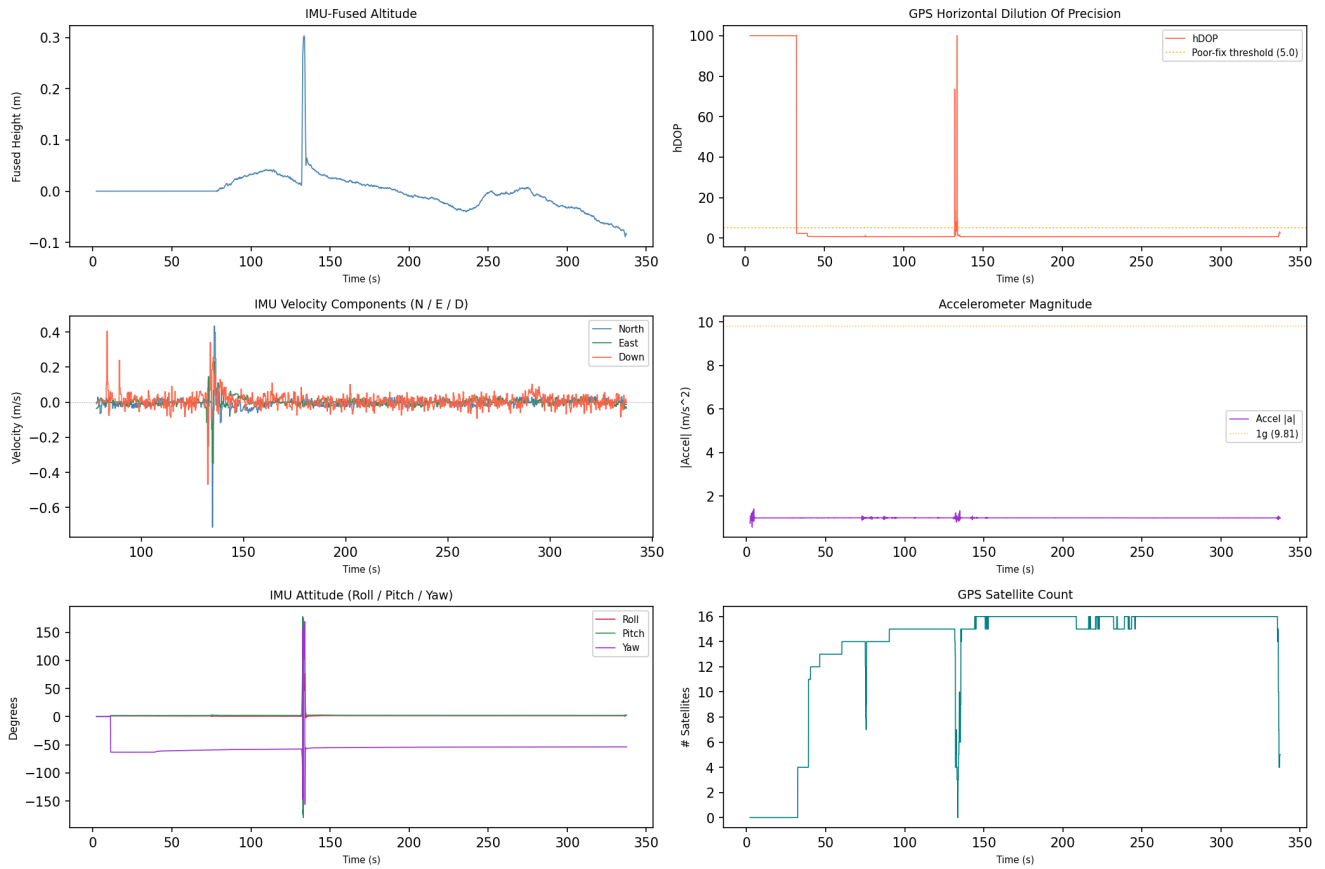
Possibly correlated (indirect indicators):

drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:b7e321ef952c4d417123970a2208663fc71b72b0b2c52068bd5bbc8cf3a511d8	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:8c0130901c10db776bb66aba8e3419a628d21c62bb208fbd5b02663ee668d6	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:df988a711915ffb87a43de6877693c062dbf40d2549782d7291939fb2ad44d48	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:5833218757f0835eb44ffafe533672ec0259301d097d018a6502befc59bee2f3	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:5ea89532ab3469120d74d3032d2564e63ad8b6ee42b02ce221c37d7072b256e6	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:e1d425f6213ecc84f6a68c13a8261d21acfd4d29bda3f74619d77973de0f7cff	
drone_sd:videos	GPS within flight bounding box (lat 39.9584, lon -106.2174)
p5:a796e8517cf21920e4667a415ee359f14da3f7f07a5fdd474d45b5f76c738514	

drone_sd:videos	GPS within flight bounding box (lat 39.9584, lon -106.2174)
p5:96b4e551824ffaf16a34254812778a25da71b41d93af663fe14647c64d0fc656	
drone_sd:videos	GPS within flight bounding box (lat 39.9608, lon -106.2201)
p5:b394e548cdf1e89eb132beb1f1559f69de9b2e6799fd6ad7fc70d571c36c2bff	
drone_sd:videos	GPS within flight bounding box (lat 39.9608, lon -106.2201)
p5:564e6bd58d751430e1a495a44803838717aebd0402e1da991043387d97207043	
drone_sd:videos	GPS within flight bounding box (lat 39.9607, lon -106.2209)
p5:2a11dd32eb3b2da78f99ce368820648d54b3512d61de05f5bbc1c917546f7bee	
drone_sd:videos	GPS within flight bounding box (lat 39.9607, lon -106.2209)
p5:2df4a3a5569fb636b7b982eed5386d52221e18c14cbd36b6dca1a70f8fbdeb2f	

Drone Log - Sensor Telemetry

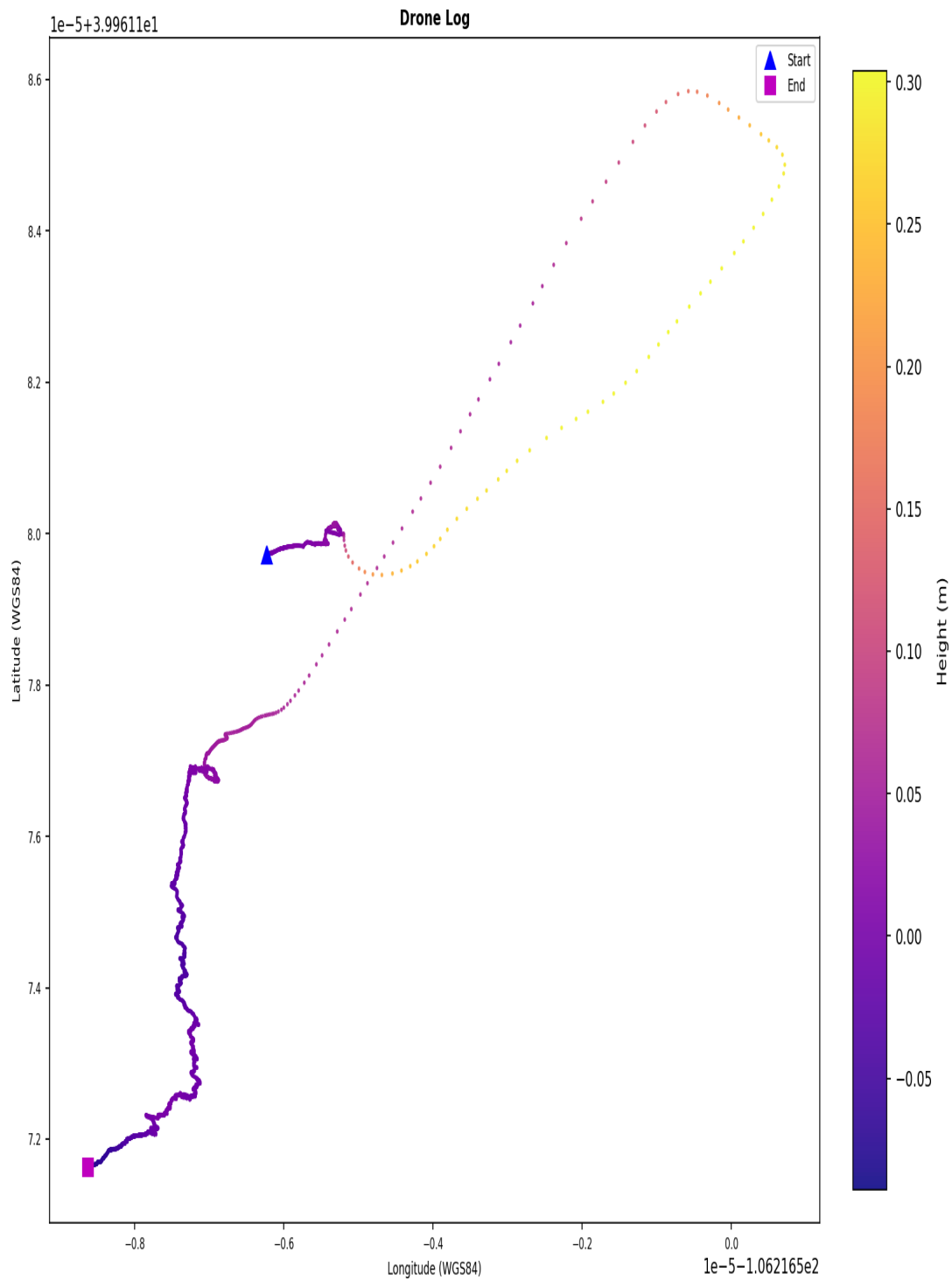
Drone Log Telemetry - flight_03



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_03



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY005.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY005.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 04

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 9183bc9d4f904893f1a386ab8c7275475615f80a46cf888d66a542ea8c3276fd
Drone Log Name: FLY006.csv
Start: 2017-06-29T18:24:47.273000+00:00
End: 2017-06-29T18:30:07.551000+00:00
Duration: 320.3 s

Flight Metrics

Peak Height: 1.0282869913420136 m
Distance Travelled: 9.368021331266068 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

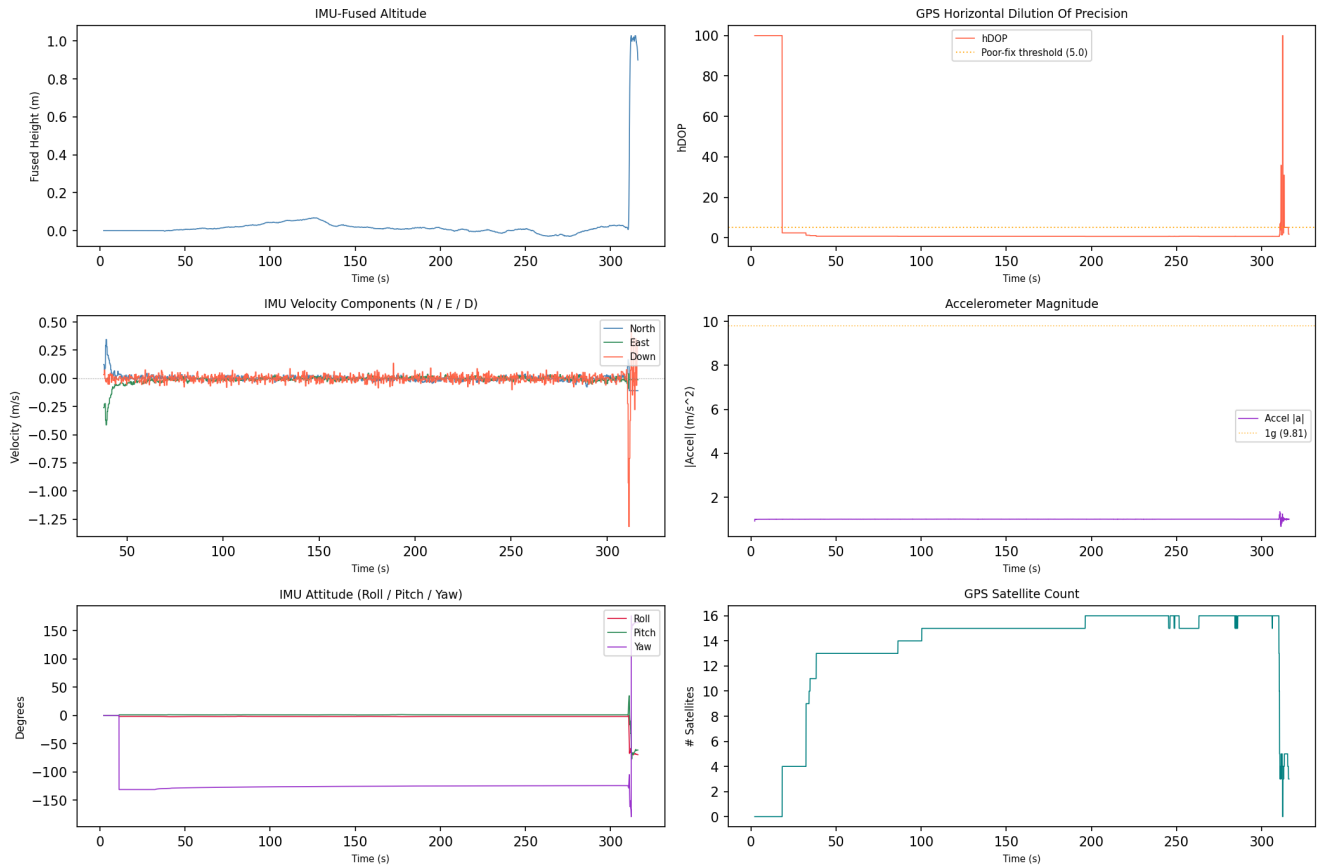
1. Manual
2. Atti
3. GPS_Atti

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2017-06-29 18:25:10	drone_flight_storage:dl	Log started	P4P
2017-06-29 18:30:03	drone_flight_storage:dl	Reached peak height	1.0 m
2017-06-29 18:30:07	drone_flight_storage:dl	Log ended	

Drone Log - Sensor Telemetry

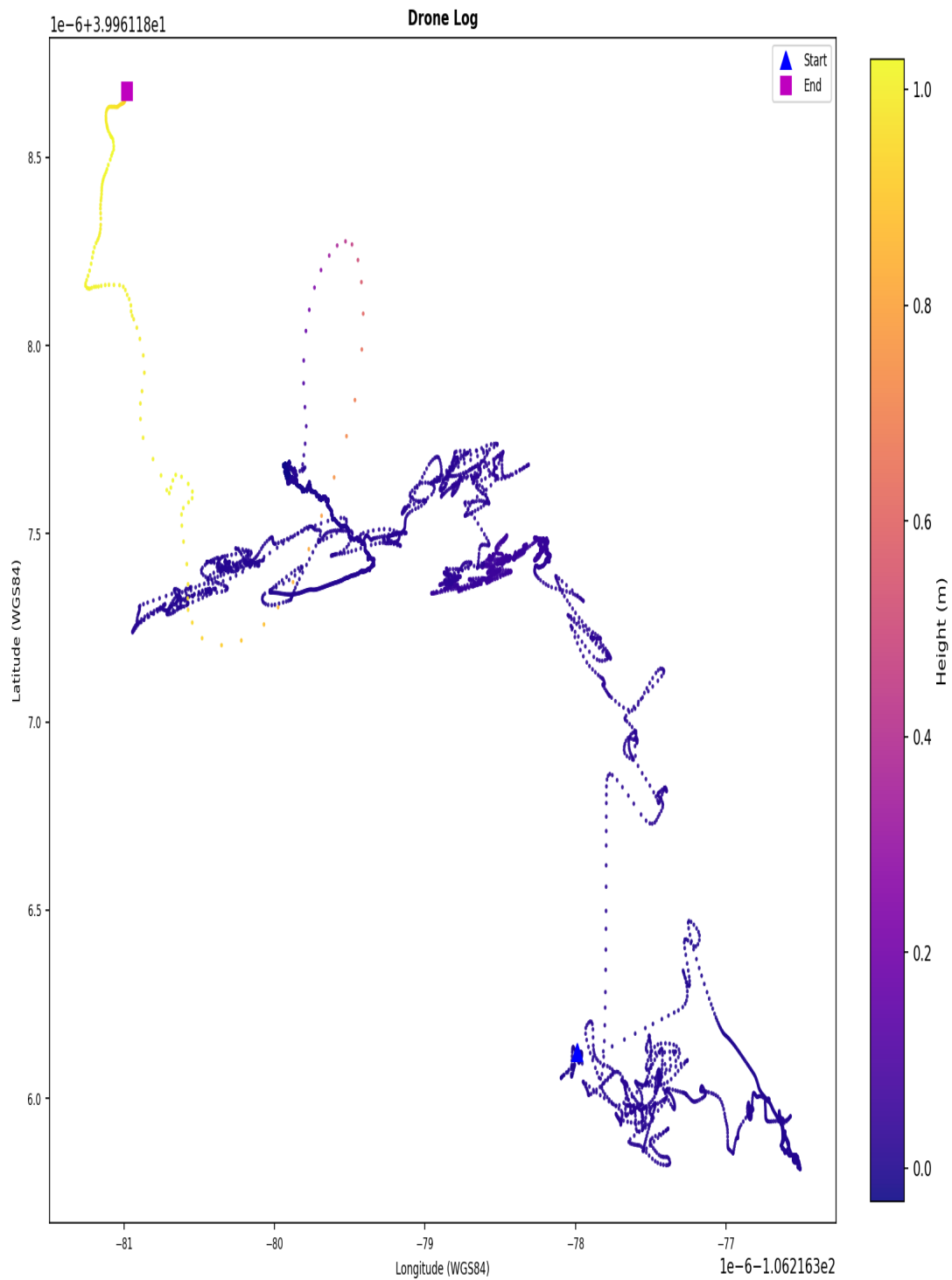
Drone Log Telemetry - flight_04



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of ± 1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_04



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: FLY006.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY006.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 05

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 77160f1321cc9fdf29ff41d000d57d1b6af3d05b52927da27edff23c9c53b74d
Drone Log Name: FLY007.csv
Start: 2017-06-29T18:38:41.869000+00:00
End: 2017-06-29T18:51:30.881000+00:00
Duration: 769.0 s

Flight Metrics

Peak Height: 120.57006692179861 m
Distance Travelled: 1715.926248983839 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AutoTakeoff
6. AssitedTakeoff

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2017-06-29 18:39:10	drone_flight_storage:dl	Log started	P4P
2017-06-29 18:39:23	drone_flight_storage:dl	Motor turned on	on
2017-06-29 18:48:44	drone_flight_storage:dl	Reached peak height	120.6 m
2017-06-29 18:51:21	drone_flight_storage:dl	Motor turned off	off
2017-06-29 18:51:21	drone_flight_storage:dl	Motor turned on	on
2017-06-29 18:51:23	drone_flight_storage:dl	Motor turned off	off
2017-06-29 18:51:31	drone_flight_storage:dl	Log ended	

Correlated Artefacts

Possibly correlated (indirect indicators):

drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:b7e321ef952c4d417123970a2208663fc71b72b0b2c52068bd5bbc8cf3a511d8	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:8c0130901c10db776bb66aba8e3419a628d21c62bb208fbd5b5d02663ee668d6	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:df988a711915ffb87a43de6877693c062dbf40d2549782d7291939fb2ad44d48	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:5833218757f0835eb44ffafe533672ec0259301d097d018a6502befc59bee2f3	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:5ea89532ab3469120d74d3032d2564e63ad8b6ee42b02ce221c37d7072b256e6	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:e1d425f6213ecc84f6a68c13a8261d21acfd4d29bda3f74619d77973de0f7cff	
drone_sd:videos	GPS within flight bounding box (lat 39.9584, lon -106.2174)

p5:a796e8517cf21920e4667a415ee359f14da3f7f07a5fdda74d45b5f76c738514

drone_sd:videos	GPS within flight bounding box (lat 39.9584, lon -106.2174)
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p5:96b4e551824ffaf16a34254812778a25da71b41d93af663fe14647c64d0fc656

drone_sd:videos	GPS within flight bounding box (lat 39.9608, lon -106.2201)
------------------------	--

p5:b394e548cdf1e89eb132beb1f1559f69de9b2e6799fd6ad7fc70d571c36c2bff

drone_sd:videos	GPS within flight bounding box (lat 39.9608, lon -106.2201)
------------------------	--

p5:564e6bd58d751430e1a495a44803838717aebd0402e1da991043387d97207043

drone_sd:videos	GPS within flight bounding box (lat 39.9607, lon -106.2209)
------------------------	--

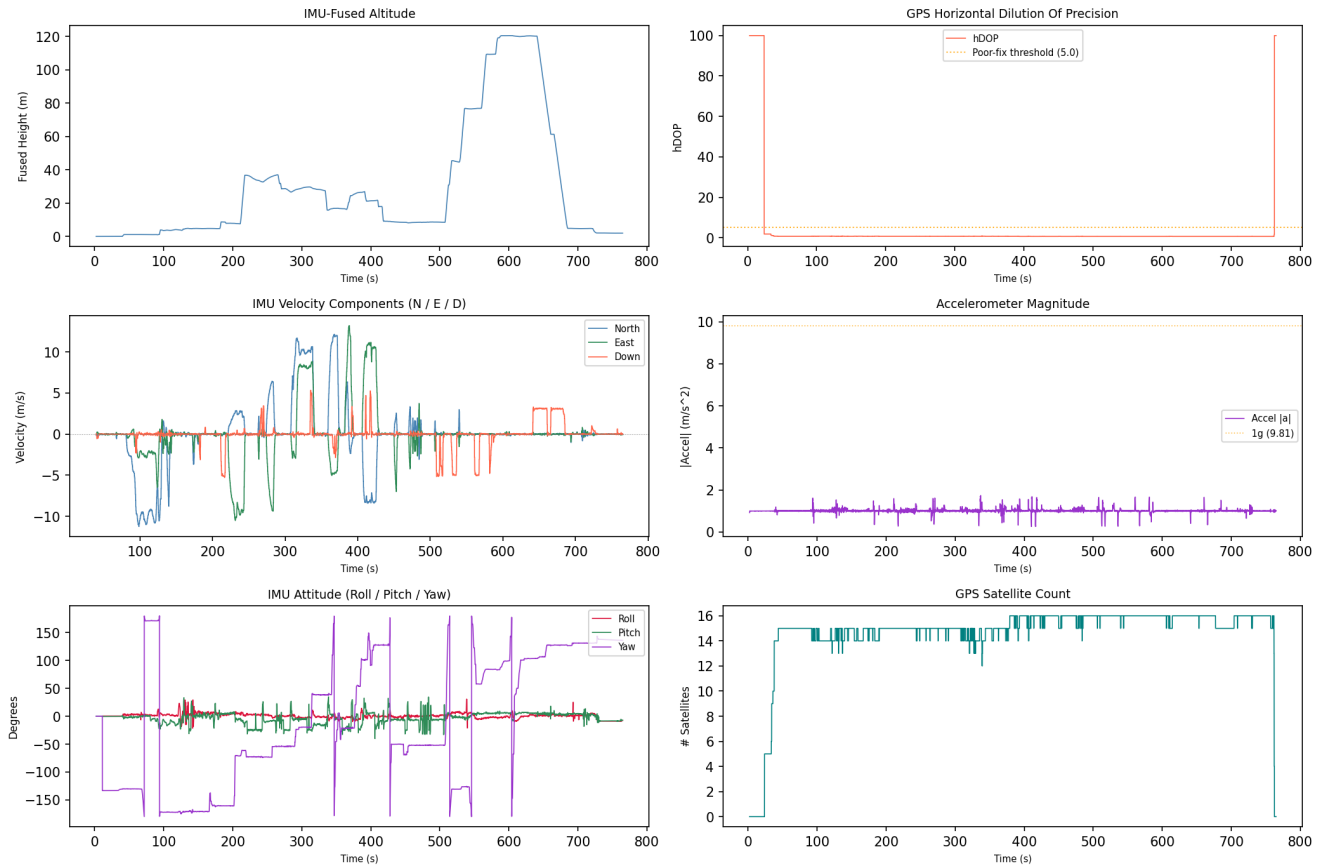
p5:2a11dd32eb3b2da78f99ce368820648d54b3512d61de05f5bbc1c917546f7bee

drone_sd:videos	GPS within flight bounding box (lat 39.9607, lon -106.2209)
------------------------	--

p5:2df4a3a5569fb636b7b982eed5386d52221e18c14cbd36b6dca1a70f8fbdeb2f

Drone Log - Sensor Telemetry

Drone Log Telemetry - flight_05



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

4. Flight Analysis - Flight 06

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Flight Record SHA-256: 4491905a98fdee1fb39dd3facd69c20ac7107e577ce65e8d70708708c13f6ab2
Flight Record Name: DJIFlightRecord_2017-06-29__13-00-45_.csv
Start: 2017-06-29T19:00:45.324000+00:00
End: 2017-06-29T19:13:38.368000+00:00
Duration: 773.0 s

Flight Metrics

Peak Height: 30.2 m
Distance Travelled: 423.89 m
Video Recording: Yes
Photo Capture: No
Photos Taken: 0
Recording Duration: 209.0 s
Record Complete: Yes

Fly Modes Observed (Chronological)

1. EngineStart
2. AutoTakeoff
3. GPS_Atti
4. Joystick
5. NaviSubMode_Tracking
6. NaviSubMode_Pointing
7. TripodGPS
8. GoHome
9. AutoLanding
10. ForceLanding

Key Events

Timestamp (UTC)	Source	Event	Details
2017-06-29 19:00:45	ctrl_ios:fr (v4.1.3)	Log started	SN: 0BMLE5501503RX; App: 4.1.3; Spark
2017-06-29 19:00:45	ctrl_ios:fr (v4.1.3)	Motor turned on	on
2017-06-29 19:00:46	ctrl_ios:fr (v4.1.3)	Record mode changed	No
2017-06-29 19:03:49	ctrl_ios:fr (v4.1.3)	Record mode changed	Starting
2017-06-29 19:04:35	ctrl_ios:fr (v4.1.3)	Record mode changed	No
2017-06-29 19:08:16	ctrl_ios:fr (v4.1.3)	Record mode changed	Starting
2017-06-29 19:08:28	ctrl_ios:fr (v4.1.3)	Record mode changed	No
2017-06-29 19:10:27	ctrl_ios:fr (v4.1.3)	Record mode changed	Starting
2017-06-29 19:10:39	ctrl_ios:fr (v4.1.3)	Record mode changed	Stop
2017-06-29 19:10:40	ctrl_ios:fr (v4.1.3)	Record mode changed	No
2017-06-29 19:10:48	ctrl_ios:fr (v4.1.3)	Record mode changed	Starting
2017-06-29 19:11:24	ctrl_ios:fr (v4.1.3)	Record mode changed	No
2017-06-29 19:11:48	ctrl_ios:fr (v4.1.3)	Reached peak height	30.2 m
2017-06-29 19:11:55	ctrl_ios:fr (v4.1.3)	Record mode changed	Starting
2017-06-29 19:13:38	ctrl_ios:fr (v4.1.3)	Log ended	Dist home: 1.2 m; Height: -4.4 m; Rec: 209.0 s

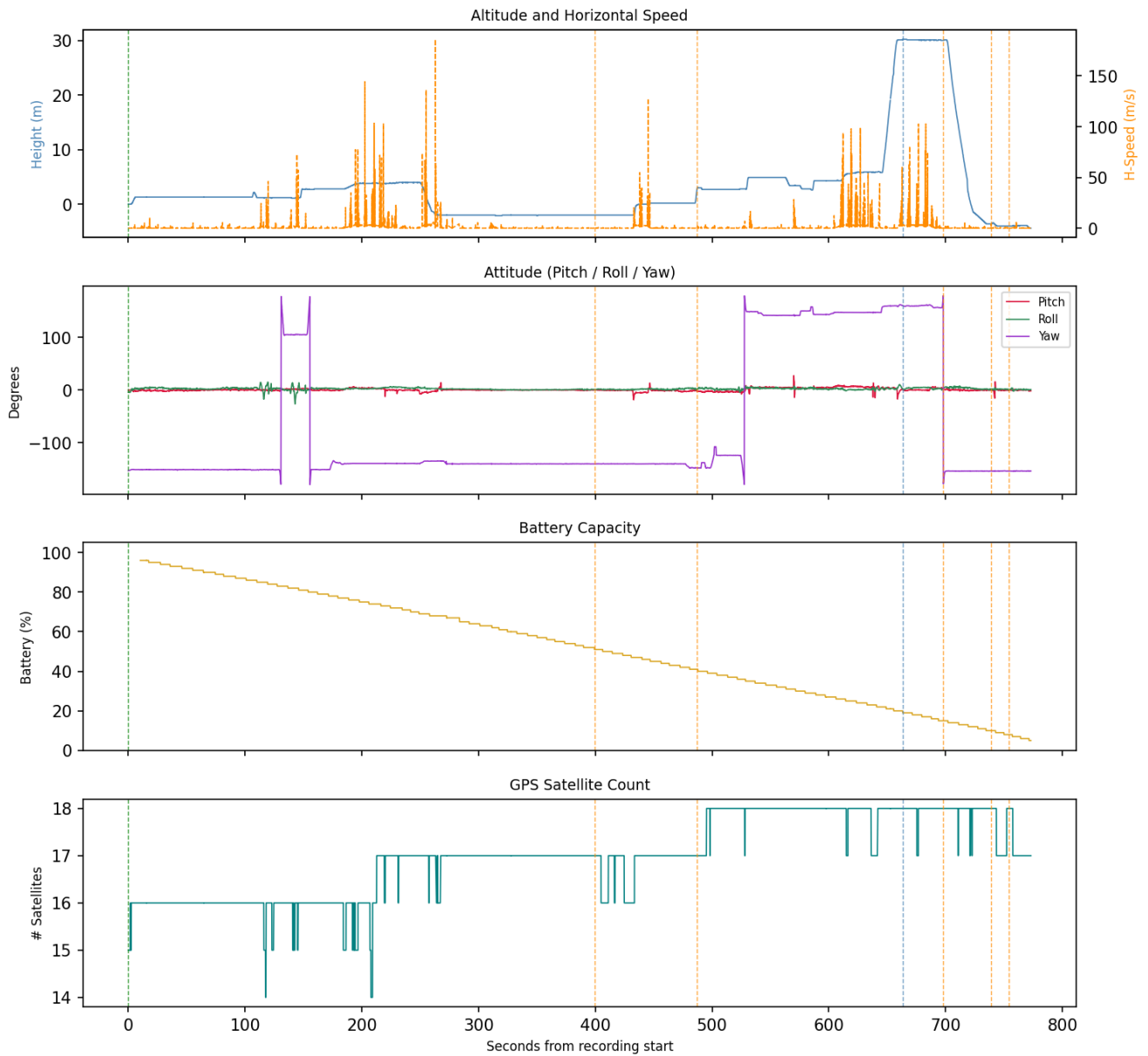
In-Flight Warnings

Timestamp (UTC)	Source	Warning Message
2017-06-29 19:07:25	ctrl_ios:fr (v4.1.3)	Warning:Tap to choose a valid zone.

Timestamp (UTC)	Source	Warning Message
2017-06-29 19:08:52	ctrl_ios:fr (v4.1.3)	Warning:Subject lost. Searching ...
2017-06-29 19:12:23	ctrl_ios:fr (v4.1.3)	Warning:The remaining battery is only enough for RTH. Return home now.
2017-06-29 19:13:04	ctrl_ios:fr (v4.1.3)	Warning:The remaining battery is only enough for RTH. Return home now.
2017-06-29 19:13:19	ctrl_ios:fr (v4.1.3)	Warning:Command Failed

Flight Record - Sensor Telemetry

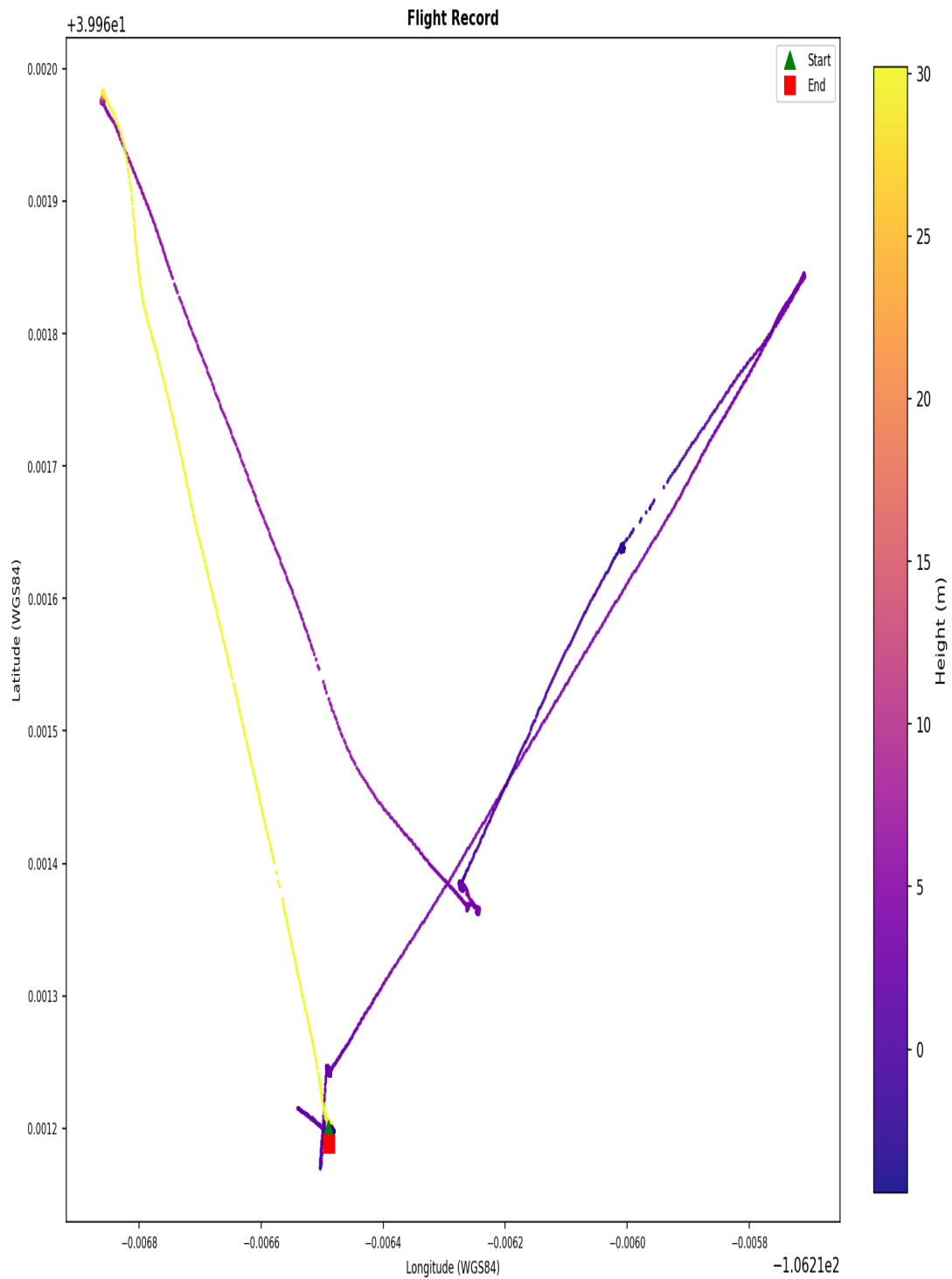
Flight Record Telemetry - flight_06



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track

Flight Ground Track - flight_06



Coordinates are WGS84. No base map. Colour represents relative height (m).

Coordinates are WGS84. No base map. Colour represents relative height (metres).

4. Flight Analysis - Flight 07

Correlation and Flight Window

Status: Unmatched (no drone log corroboration)
Drone Log SHA-256: 69ddbdf905e27dbf84c47afcaff5eec7cf5cb3f4ccd58951d8e2206874310721
Drone Log Name: FLY008.csv
Start: 2017-06-29T20:26:17.019000+00:00
End: 2017-06-29T20:49:56.669000+00:00
Duration: 1419.7 s

Flight Metrics

Peak Height: 115.6851035468051 m
Distance Travelled: 2264.5885478447535 m
Video Recording: No
Photo Capture: No

Fly Modes Observed (Chronological)

1. Manual
2. Atti
3. GPS_Atti
4. ASST_TAKEOFF
5. AutoTakeoff

Key Events

Timestamp (UTC)	Source	Event	Details
-	drone_flight_storage:dl	Motor turned off	off
2017-06-29 20:26:55	drone_flight_storage:dl	Log started	P4P
2017-06-29 20:35:16	drone_flight_storage:dl	Motor turned on	on
2017-06-29 20:46:56	drone_flight_storage:dl	Reached peak height	115.7 m
2017-06-29 20:49:12	drone_flight_storage:dl	Motor turned off	off
2017-06-29 20:49:57	drone_flight_storage:dl	Log ended	

Correlated Artefacts

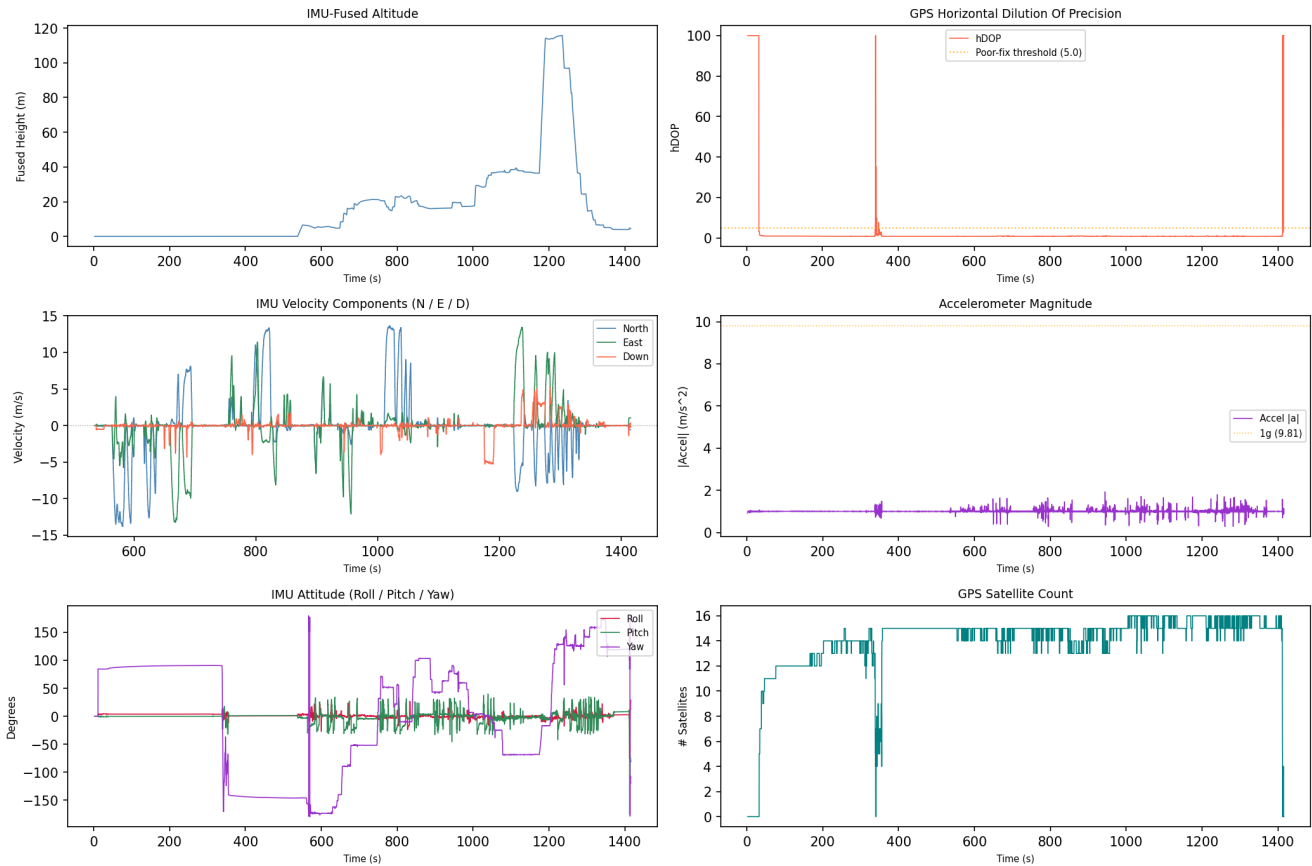
Possibly correlated (indirect indicators):

drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:b7e321ef952c4d417123970a2208663fc71b72b0b2c52068bd5bbc8cf3a511d8	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:8c0130901c10db776bb66aba8e3419a628d21c62bb208fbd5bdf02663ee668d6	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:df988a711915ffb87a43de6877693c062dbf40d2549782d7291939fb2ad44d48	
drone_sd:videos	GPS within flight bounding box (lat 39.9575, lon -106.2177)
p5:5833218757f0835eb44ffafe533672ec0259301d097d018a6502befc59bee2f3	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:5ea89532ab3469120d74d3032d2564e63ad8b6ee42b02ce221c37d7072b256e6	
drone_sd:videos	GPS within flight bounding box (lat 39.9612, lon -106.2165)
p5:e1d425f6213ecc84f6a68c13a8261d21acfd4d29bda3f74619d77973de0f7cff	
drone_sd:videos	GPS within flight bounding box (lat 39.9584, lon -106.2174)
p5:a796e8517cf21920e4667a415ee359f14da3f7f07a5fdda74d45b5f76c738514	
drone_sd:videos	GPS within flight bounding box (lat 39.9584, lon -106.2174)
p5:96b4e551824ffaf16a34254812778a25da71b41d93af663fe14647c64d0fc656	

drone_sd:videos	GPS within flight bounding box (lat 39.9608, lon -106.2201)
p5:b394e548cdf1e89eb132beb1f1559f69de9b2e6799fd6ad7fc70d571c36c2bff	
drone_sd:videos	GPS within flight bounding box (lat 39.9608, lon -106.2201)
p5:564e6bd58d751430e1a495a44803838717aebd0402e1da991043387d97207043	
drone_sd:videos	GPS within flight bounding box (lat 39.9607, lon -106.2209)
p5:2a11dd32eb3b2da78f99ce368820648d54b3512d61de05f5bbc1c917546f7bee	
drone_sd:videos	GPS within flight bounding box (lat 39.9607, lon -106.2209)
p5:2df4a3a5569fb636b7b982eed5386d52221e18c14cbd36b6dca1a70f8fbdeb2f	

Drone Log - Sensor Telemetry

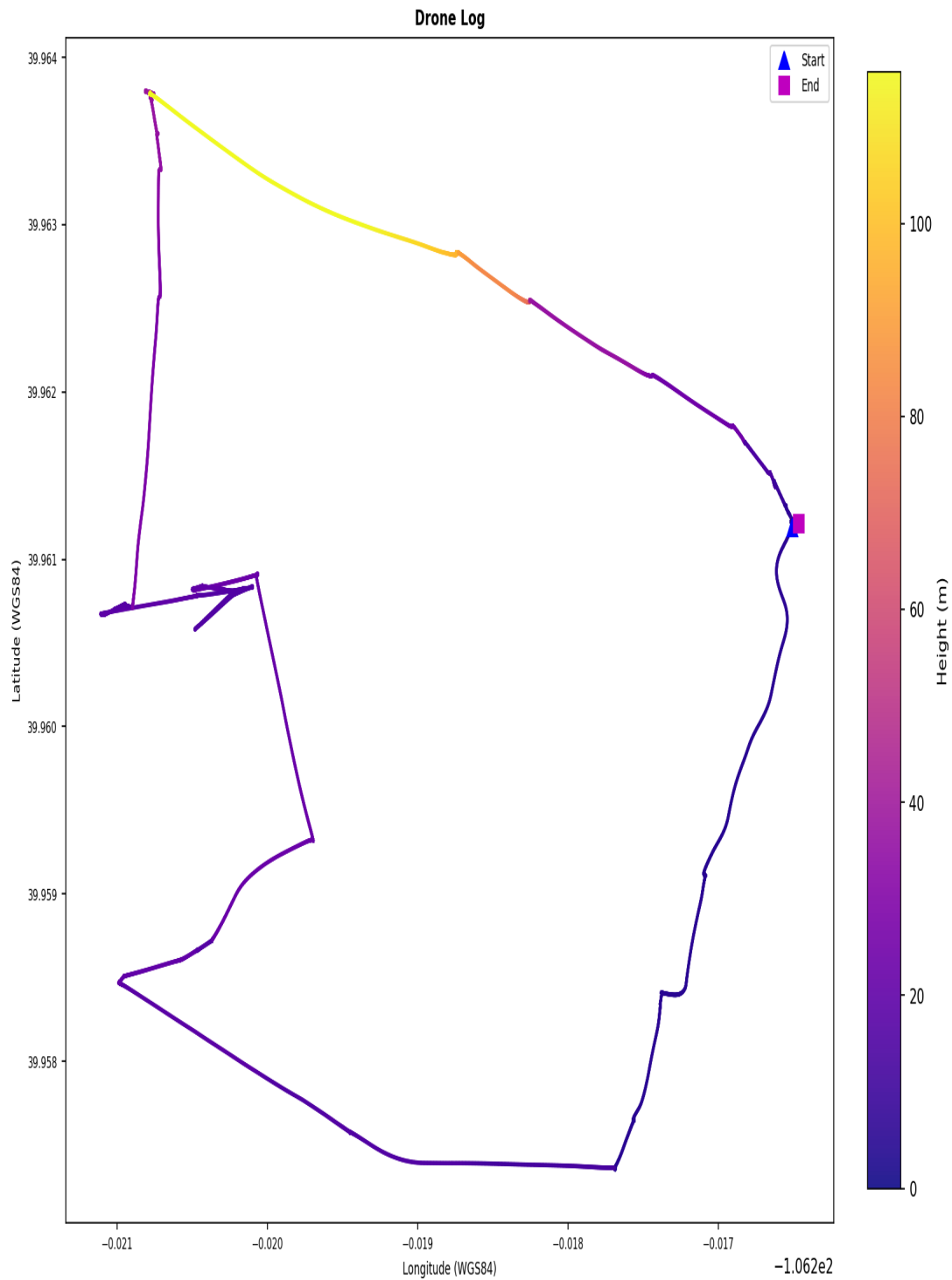
Drone Log Telemetry - flight_07



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track

Flight Ground Track - flight_07



Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (FLY008.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path. Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	No
controller_ios	Yes
drone_sd	Yes
drone_flight_storage	Yes

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_ios	2	12	0	3	1	7	7
drone_sd	0	0	0	0	0	17	12
drone_flight_storage	0	0	11	0	0	0	0

Data Quality Notes per Category

Databases (12 files):

- 3 database(s) empty (controller_ios)

Drone Logs (11 files):

- 11 file(s) with row anomalies
- 11 file(s) with empty columns
- 11 file(s) with constant-value columns

Flight Logs (3 files):

- 1 file(s) unreadable (binary)
- formats: 1 bracketed_log, 1 json_array_log

Flight Records (1 files):

- 1 file(s) with row anomalies
- 1 file(s) with empty columns
- 1 file(s) with constant-value columns

Images (24 files):

- 17 file(s) contain(s) EXIF GPS coordinates
- 24 file(s) missing (normalisable) EXIF timestamp
- 7 file(s) missing EXIF GPS coordinates

Videos (19 files):

- 12 file(s) contain(s) EXIF GPS coordinates
- 19 file(s) missing (normalisable) EXIF timestamp
- 7 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Drone_logs: 11 file(s) with row anomalies detected in telemetry data.
- Drone_logs: 11 file(s) with empty columns - some telemetry channels contain no data.
- Drone_logs: 11 file(s) with constant-value columns - some telemetry channels may be unreliable.
- Flight_logs: 1 file(s) unreadable (binary) - file content could not be parsed.
- Flight_logs: log files present in the following format(s): 1 bracketed_log, 1 json_array_log.
- Flight_records: 1 file(s) with row anomalies detected in telemetry data.
- Flight_records: 1 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 1 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p3 - controller android]: source evidence not found
3. [p3 - controller ios - drone logs]: no artefacts found
4. [p4 - controller android]: source evidence not found
5. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY000.csv
6. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY001.csv
7. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY004.csv
8. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY009.csv
9. [p6 - drone flight storage - drone logs]: no parseable timestamps in drone log: norm_FLY010.csv

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Databases

djiFMDB.db

- database_empty: all tables empty

djiFMDB_1.db

- database_empty: all tables empty

statistics.db

- database_empty: all tables empty

Drone Logs

norm_FLY000.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 5 row(s) affected
- zero_coordinate: 272 row(s) affected
- gps_poor_fix: 272 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 133 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+128 more)
- gps_accuracy_poor: 272 row(s) affected

norm_FLY001.csv

- missing_gps: 15 row(s) affected
- zero_coordinate: 1845 row(s) affected
- gps_poor_fix: 14467 row(s) affected
- contains_no_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains_constant_value: 98 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+93 more)
- gps_accuracy_poor: 16078 row(s) affected
- quaternion_invalid: 244 row(s) affected

norm_FLY002.csv

- missing_gps: 7 row(s) affected
- zero_coordinate: 1984 row(s) affected
- gps_poor_fix: 16196 row(s) affected
- contains_no_value: 18 column(s) - 4.3.0, ConvertDatV3, IMUCalcs(0):Lat:C, IMUCalcs(0):Long:C, IMUCalcs(0):diffVelD:C (+13 more)
- contains_constant_value: 91 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+86 more)
- gps_accuracy_poor: 17185 row(s) affected
- quaternion_invalid: 253 row(s) affected

norm_FLY003.csv

- missing_gps: 8 row(s) affected
- gps_poor_fix: 4726 row(s) affected
- contains_no_value: 18 column(s) - 4.3.0, ConvertDatV3, IMUCalcs(0):Lat:C, IMUCalcs(0):Long:C, IMUCalcs(0):diffVelD:C (+13 more)
- contains_constant_value: 126 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+121 more)
- gps_accuracy_poor: 5116 row(s) affected
- quaternion_invalid: 79 row(s) affected

norm_FLY004.csv

- missing_gps: 16 row(s) affected
- zero_coordinate: 3635 row(s) affected
- gps_poor_fix: 6595 row(s) affected
- contains_no_value: 25 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+20 more)
- contains_constant_value: 108 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+103 more)
- gps_accuracy_poor: 6595 row(s) affected
- quaternion_invalid: 234 row(s) affected

norm_FLY005.csv

- missing_gps: 18 row(s) affected
- zero_coordinate: 781 row(s) affected
- gps_poor_fix: 861 row(s) affected
- contains_no_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, osd_data:lowVoltage
- contains_constant_value: 74 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+69 more)
- gps_accuracy_poor: 874 row(s) affected
- quaternion_invalid: 248 row(s) affected
- attitude_out_of_bounds: 41 row(s) affected

norm_FLY006.csv

- missing_gps: 7 row(s) affected
- zero_coordinate: 451 row(s) affected
- motor_airborne_off: 82 row(s) affected
- gps_poor_fix: 485 row(s) affected
- contains_no_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, osd_data:lowVoltage
- contains_constant_value: 88 column(s) - ATTI_MINIO:s_rsv00, ATTI_MINIO:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+83 more)
- gps_accuracy_poor: 555 row(s) affected
- quaternion_invalid: 249 row(s) affected

norm_FLY007.csv

- missing_gps: 15 row(s) affected
- zero_coordinate: 597 row(s) affected
- motor_airborne_off: 229 row(s) affected
- gps_poor_fix: 669 row(s) affected
- contains_no_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C,

RC_Info:dataLost (+1 more)

- contains_constant_value: 55 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D, BatteryInfo:out_ctl_f:D (+50 more)
- gps_accuracy_poor: 669 row(s) affected
- quaternion_invalid: 244 row(s) affected

norm_FLY008.csv

- missing_gps: 14 row(s) affected
- zero_coordinate: 432 row(s) affected
- motor_airborne_off: 1279 row(s) affected
- gps_poor_fix: 947 row(s) affected
- contains_no_value: 5 column(s) - 4.3.0, ConvertDatV3, IMU_ATT(0):absoluteHeight:C, IMU_ATT(0):distanceHP:C, RC_Info:dataLost
- contains_constant_value: 55 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:dyna_cnt:D, BatteryInfo:out_ctl:D (+50 more)
- gps_accuracy_poor: 998 row(s) affected
- quaternion_invalid: 240 row(s) affected
- attitude_out_of_bounds: 26 row(s) affected

norm_FLY009.csv

- missing_gps: 11 row(s) affected
- zero_coordinate: 290 row(s) affected
- gps_poor_fix: 290 row(s) affected
- contains_no_value: 26 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+21 more)
- contains_constant_value: 118 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+113 more)
- gps_accuracy_poor: 290 row(s) affected
- quaternion_invalid: 246 row(s) affected

norm_FLY010.csv

- missing_gps: 12 row(s) affected
- zero_coordinate: 10853 row(s) affected
- gps_poor_fix: 10853 row(s) affected
- contains_no_value: 24 column(s) - 4.3.0, ConvertDatV3, GPS:Date, GPS:Time, GPS:dateTimeStamp (+19 more)
- contains_constant_value: 103 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:MotorSpd:D, AirSpeed:air_vbx:D, AirSpeed:air_vby:D (+98 more)
- gps_accuracy_poor: 10853 row(s) affected
- quaternion_invalid: 251 row(s) affected
- attitude_out_of_bounds: 22 row(s) affected

Flight Logs

2017-06-29_13_00_45.dat

- format: json_array_log | entries: 30

upgradeLog_0.txt

- format: bracketed_log | entries: 52

Flight Records

norm_DJIFlightRecord_2017-06-29_13-00-45.csv

- altitude_negative: 2119 row(s) affected
- altitude_spike: 27 row(s) affected
- contains_no_value: 59 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, APP_SER_WARN.warn, CENTER_BATTERY.connStatus (+54 more)
- contains_constant_value: 141 column(s) - CAMERA_INFO.cameraType, CAMERA_INFO.connectState, CAMERA_INFO.enabledPhoto, CAMERA_INFO.encryptStatus, CAMERA_INFO.firmErrorType (+136 more)

Images

2017_06_29_13_03_48.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_13_08_15.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_13_10_27.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_13_10_48.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_13_11_55.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_13_13_49.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_13_13_49.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0004.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0007.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0008.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0010.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0012.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0013.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0014.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0015.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0016.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0017.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0018.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0019.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0020.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0021.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0022.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0023.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

Videos

2017_06_29_13_03_48.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_13_08_15.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_13_10_27.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_13_10_48.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_13_11_55.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_13_13_04.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2017_06_29_13_13_49.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0002.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0003_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0003.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0005_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0005.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0006_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0006.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0009_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0009.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0011_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0011.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Drone Model inconsistency detected - conflicting values observed: P4P, Spark. Manual verification required.
2. DJI Application Version inconsistency detected - conflicting values observed: 3.1.12, 4.1.3. Manual verification required.
3. Product Type inconsistency detected - conflicting values observed: 2, 26. Manual verification required.
4. Note: no firmware version found - Device and account information.
5. Last App Launch inconsistency detected - conflicting values observed: 2017-06-28T19:55:11Z, 2017-06-29T16:40:17Z. Manual verification required.
6. Note: no last known latitude (find aircraft) found - Device and account information.
7. Note: no last known longitude (find aircraft) found - Device and account information.
8. Note: no last known altitude (find aircraft) found - Device and account information.
9. Note: no last known heading (find aircraft) found - Device and account information.
10. Note: no last known horiz. accuracy (find aircraft) found - Device and account information.
11. 7 flight record(s) could not be correlated to a drone log (flight_01, flight_02, flight_03, flight_04, flight_05, flight_06, flight_07). Independent corroboration is absent for these flight(s).
12. 9 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

com.dji.go.plist		Account Data	controller_ios	extract_logical
SHA-256:	5fa7e27fc57df3de9955d1f01b00b5f7aa511d3af8bdebfb233a74e4f45950b8			
SHA-1:	b791716fa4b79857b6ea20f35129470ab9181a7c			

com.dji.pilot.plist		Account Data	controller_ios	extract_logical
SHA-256:	205a9a279440e001bf2d523ac4f59a52f0cbd9a48c859845be3bf6809ac43ff4			
SHA-1:	de721e27e6982111d5e457eb6117d04075cbacbb			

airmap_flysafeplaces.db		Databases	controller_ios	extract_logical
SHA-256:	d02f441ba79d6c82b33db8c2eca742ffc278bb998eb46edaa57290f53b992ee6			
SHA-1:	805373eb7dd57a932cc3438234763f1a27b61a9c			

airmap_flysafeplaces_1.db		Databases	controller_ios	extract_logical
SHA-256:	01472f913caf3ce5d50b146cd37a6fea1a18ca4d9666b2d5dc06189a8ac5c17c			
SHA-1:	fdecfca00525fd0f36dc51b5a8709efaf409f908			

djiFMDB.db		Databases	controller_ios	extract_logical
SHA-256:	baa7277998f1c5db44be2b44a2264bedd94f93495fbb0d73636ee2fdc9c94848			
SHA-1:	d6720f20112c173533a7ad8ccedd8c8992503ddb			

djiFMDB_1.db		Databases	controller_ios	extract_logical
SHA-256:	003687bc29cf9f101fae000e55fef1718de111f65242f517487ab2cb826a3e37			
SHA-1:	86fe40031cfeab9bdb5627a6b0bb7a446abb05f1			

nfz_flysafeplaces.db		Databases	controller_ios	extract_logical
SHA-256:	e82bd13f41138afd8f5f7fe1a0343a0b20bbcb4bfab48ad7936899c0b666420b7			
SHA-1:	2b47e2d280ab7baced2fd2d270da5ac6c9935629			

nfz_flysafeplaces_1.db		Databases	controller_ios	extract_logical
SHA-256:	7b9f33fc9adef204358cf4ae773c8ab650f17238d78df654f361a2e68589981c			
SHA-1:	295aee5b07fff976495f0fda92350e0e4eaa0e88			

statistics.db		Databases	controller_ios	extract_logical
SHA-256:	83bfc672a35016dcdffa168cbc780d6219345ddfe601ad75087ce879503a3058a			
SHA-1:	3a8843449db0f78d4502b81d67e0f26584226a80			

statistics_1.db		Databases	controller_ios	extract_logical
SHA-256:	d56993b9997271afe152dc55524cf2a77000c26fbd245a962e677996a0cffb2a			
SHA-1:	769dc74f506cc7c7b5cec1fb696295184925af5c			

tencent_analysis.db		Databases	controller_ios	extract_logical
SHA-256:	4d09741e05365fc64335fade02f229f4e9627f58d0a9063368a0a1e078b1290f			
SHA-1:	41367b5380af044fa85d582ef938b3be64bc688c			

tencent_analysis_1.db		Databases	controller_ios	extract_logical
SHA-256:	f2afa9f4d14bd8dbc4fa0c523b9e70cca956e15172aee22457eb5e1a52508353			
SHA-1:	d95cb10e47a20cad57f393b79cc8690f9673aef9			

tencent_wxo_analysis.db	Databases	controller_ios	extract_logical
SHA-256:	f87c236548671234f6491a57927a5f5dfa0e7d48c36d8bc5a7fd669f68a5419d		
SHA-1:	f09321eaafc8cd2ec99c6af6e995773b3422fd7e		

tencent_wxo_analysis_1.db	Databases	controller_ios	extract_logical
SHA-256:	ebff694aafb59abcc097ddce8ceb2006ae865e74ef7bd299c604964f9f94d28		
SHA-1:	0d8f20ebdf2dcb5e4564a65cf51e2fceca70b350		

FLY000.DAT	Drone Logs	drone_flight_storage	extract_physical
SHA-256:	40910fa3e293ceff72384866cefc2d8cfe445b369108eb650ee253fa34756d62		
SHA-1:	a1a6b5437d8be2c57fb96d9f17c4409b719dea5a		

FLY001.DAT	Drone Logs	drone_flight_storage	extract_physical
SHA-256:	3b9e1db78a6933c0b6647abe6b1367de45226050a6d13a0ae237772746b80dd3		
SHA-1:	66b20d6cdf2aaffd82e8e8a27a6f89f4e014d67b		

FLY002.DAT	Drone Logs	drone_flight_storage	extract_physical
SHA-256:	078e6f73bb39451672e0550eb1f0b266abd2eabed6ba13bb38ffd58d73194afe		
SHA-1:	b4be20bee5159cffaf27762f0c43ec591f340f53		

FLY003.DAT	Drone Logs	drone_flight_storage	extract_physical
SHA-256:	8f67c39f42b98b3e27ac5b72eef129e59a9db0602aa0a95a99f3579ca7f944ca		
SHA-1:	16f0b9b93e1d3ebae90e914743347ef2bc722523		

FLY004.DAT	Drone Logs	drone_flight_storage	extract_physical
SHA-256:	2202fdaf6407238f1bb1a3656dc1e41244c6a06ce59242f877564015ef82e360		
SHA-1:	6899942028c02d83021284560f4900179a84fdcc		

FLY005.DAT	Drone Logs	drone_flight_storage	extract_physical
SHA-256:	c139fdc624b47bd2d6caa8d9626aeaa3bcf64e7e43ded05bddcf5724a9c4785b		
SHA-1:	c1514bc1487795d7f36c30ca41191b44ced4ca2d		

FLY006.DAT	Drone Logs	drone_flight_storage	extract_physical
SHA-256:	8efcb686b413ecac6d39b10502c299ffb32a0173263eaae6a918faa04eb1c757		
SHA-1:	d6967f4fc5e9ef15e62b1c652ced2859876e3159		

FLY007.DAT	Drone Logs	drone_flight_storage	extract_physical
SHA-256:	34efb6cb1096929488198b989e69f93447f57862ba6a2454c058e11e04c8f469		
SHA-1:	cf61cf4e317590c21ec1d6ca57c45514d7eae4ec		

FLY008.DAT	Drone Logs	drone_flight_storage	extract_physical
SHA-256:	f5b639c7336f87d170212acbe4fa43aeca30262f2593ec52e6643e5841cca741		
SHA-1:	b66b76ab3667a709252ed1887b5c2c9722dc3ed1		

FLY009.DAT	Drone Logs	drone_flight_storage	extract_physical
SHA-256:	d6708dc80052a52512279b8dc8a9faf4a5c29818d7fcb27a9995a78578642884		
SHA-1:	b5d9bd857406c01630009fdaf140e8202a2430e2		

FLY010.DAT	Drone Logs	drone_flight_storage	extract_physical
SHA-256:	6962f2dd63ede85ecad20f31c5d12fc50dbc23c3f4889a969683f961e72990ec		
SHA-1:	251f9e7d60291c3f6d0fd04743e902aad018166f		

2017-06-29_13_00_45.dat	Flight Logs	controller_ios	extract_logical
SHA-256:	1294860f721ba01f40c06d8d0240d9f71e46472cef8eebf705334e10d695cab5		
SHA-1:	d391b9b0c9637624030fe36c773ab42107610fbe		

app_log.txt	Flight Logs	controller_ios	extract_logical
SHA-256:	984b47590c61e7a77208ad525dec6ac15dc598c26240b86543560cda7145453a		
SHA-1:	8ed37ded8b1f5bac7bcac870c5a867a21abd09fd		

upgradeLog_0.txt	Flight Logs	controller_ios	extract_logical
SHA-256:	72615531d06959b848f2376566b70348e99112f170432b66782b49e50a9e2f03		
SHA-1:	d1513b5079013eeff264273990de78ba53b0c1be		

DJIFlightRecord_2017-06-29_13-00-45_.txt	Flight Records	controller_ios	extract_logical
SHA-256:	e95f139faa0531feab5beebb2663a8f864ebd449e522cc49c7764d9d656cbee2		
SHA-1:	3bb4a51ed0c2a97ebe262638726265201f5c9770		

2017_06_29_13_03_48.thumbnail	Images	controller_ios	extract_logical
SHA-256:	4c059da7399c97ccff2ff88db4c7863dba1173efda80cf691aae1b14bd8e035b		
SHA-1:	7e3c7b986cc475613ba3eee653ad1a2fc375b9fa		

2017_06_29_13_08_15.thumbnail	Images	controller_ios	extract_logical
SHA-256:	57eebc4d58558f4fbdb1dc16740462a9728c82d2d1dd7dcef2c31836ebdbf500		
SHA-1:	1638fc42f371dfea50316f61a2ead255f1b89c30		

2017_06_29_13_10_27.thumbnail	Images	controller_ios	extract_logical
SHA-256:	5d1e8e253bb2eecb432c97c333e4a1727ab44d858802227746e5093ebba59661		
SHA-1:	d01c6ce854f69d4250a9f555c0c67725eb49c832		

2017_06_29_13_10_48.thumbnail	Images	controller_ios	extract_logical
SHA-256:	0ea98faf8581e0a53f32d39f48cddf5c47062a75a46fa7af5aa592fc6afec40f		
SHA-1:	0db376d93eafd82eb3f143820ef4a4e62b080a42		

2017_06_29_13_11_55.thumbnail	Images	controller_ios	extract_logical
SHA-256:	b17227732ea0f7e6d9768f2096e16064dab68869752ccce98dccc0392de1f3376		
SHA-1:	66618ae403604a6c4e6bae7ad44093b0c7cb1584		

2017_06_29_13_13_04.thumbnail	Images	controller_ios	extract_logical
SHA-256:	a62b37253e288d9b1df0a4fbca0c1c7b85f38cd0a10acbd34f2b36dca8fe66d5		
SHA-1:	4bb76747a5fd19c9d476d9237e87e5e09e4df22a		

2017_06_29_13_13_49.thumbnail	Images	controller_ios	extract_logical
SHA-256:	a62b37253e288d9b1df0a4fbca0c1c7b85f38cd0a10acbd34f2b36dca8fe66d5		
SHA-1:	4bb76747a5fd19c9d476d9237e87e5e09e4df22a		

DJI_0001.JPG	Images	drone_sd	extract_physical
SHA-256:	ca120108a8b18e5deb186db4ff0a0783f13be9ec340a609529c550cd565b3900		
SHA-1:	2d0d18e92731266a95383b340e929e864f262a22		

DJI_0004.JPG	Images	drone_sd	extract_physical
SHA-256:	156179a4c036f4ff09313fa145042228cdcee0792ea1687dd2bfb714a472a7ea		
SHA-1:	036116b50e76e778c28d55bccd43ffc86e71c70a		

DJI_0007.JPG	Images	drone_sd	extract_physical
SHA-256:	b6ccdc3903027f542d8fd0379c4f1001f994c638a5d5030195bbc8a7a14468fc		
SHA-1:	2f151e5af1d671012da8174f2722f1a073a0a1ac		

DJI_0008.JPG	Images	drone_sd	extract_physical
SHA-256:	ba6f37f102e847211489b5371fdb817329d8555f566a132e310763588a20b20e		
SHA-1:	0df47b41bc76c63298d8170188e7628bc9b68fea		

DJI_0010.JPG	Images	drone_sd	extract_physical
SHA-256:	01d51be119db0b02d8d7b7e31a01b115411c26dfe177eb66b5dfc3de6d60868c		
SHA-1:	dca59ae3bed017f16d4bed0f2ba8ea42f5b487ec		

DJI_0012.JPG	Images	drone_sd	extract_physical
SHA-256:	aa97f908554705972fbb0362d863056286f863d4c5b17bba34190f4440f90f8b		
SHA-1:	0248de6ae99d250e905beed71f78f9f3a66ea3bb		

DJI_0013.JPG	Images	drone_sd	extract_physical
SHA-256:	079838ccc4ed59bd93bc6aaa74a337a092063749764e9ee9326df3b3f769107d		
SHA-1:	7ffab19a4dc11ff1372acf0f6511643fbb1e5f61		

DJI_0014.JPG	Images	drone_sd	extract_physical
SHA-256:	e00139d2a43b8ae05712f06bc14e6a0f82e64869b9c182f61c61afb888109422		
SHA-1:	f399030b3368c679c3d257a4068d38cf898dda50		

DJI_0015.JPG	Images	drone_sd	extract_physical
SHA-256:	0182321ece95430b6c9dbf7e27a49956dc9761c24f5cc9b03be0a9c47ad25c6d		
SHA-1:	bb27a028481511218f1b2ad4a7ddbe6fe376003a		

DJI_0016.JPG	Images	drone_sd	extract_physical
SHA-256:	74c0691b437ec9e3911cbddf5de1b6654c4f80a36e540b128dbb02f993df566e		
SHA-1:	f67054ecf668b105b931caf07e556d4246334fbc		

DJI_0017.JPG	Images	drone_sd	extract_physical
SHA-256:	f3ebdeef41ffd3b5ccf62ec7468f6f9b7d86a541bda848ecb4dd9c162f4e49e5		
SHA-1:	8f53aa4171ad2815726870c992ae09326721d426		

DJI_0018.JPG	Images	drone_sd	extract_physical
SHA-256:	54922dafa9933d3fe0fb2ae7f657b3fb1d41c9516334e19e6a1b486fd0190993		
SHA-1:	811db60956180dc3e4441a3ef10033ef35048969		

DJI_0019.JPG	Images	drone_sd	extract_physical
SHA-256:	fbc22eb8455bd1f350c4a9fc9df97770072fcc727a354d2434dc1550b0b5b3bd		
SHA-1:	088d08ef0b0b6f0a88f77d386e9251d5ff294fc7		

DJI_0020.JPG	Images	drone_sd	extract_physical
SHA-256:	8d90018ed19e66acde8b90016826607c5416c0e38414b5e567e6b19d7815aa57		
SHA-1:	6bec9dd0cbda94a0e0e9539e0449eaaa16eebc21		

DJI_0021.JPG	Images	drone_sd	extract_physical
SHA-256:	9041aae2121b0393c0d4c0ddf3e2f15b4f0df5249b4179431dd22c46046b416b		
SHA-1:	1523175ac768e5b769a5dfc45855d14a4df8ca83		

DJI_0022.JPG	Images	drone_sd	extract_physical
SHA-256:	4169ef410cc02adc6c5eb68509a75b3da17608efc19b0ab98e77813c9cd648a0		
SHA-1:	b6627886a03a9692f5c878303c7918c3025796e7		

DJI_0023.JPG	Images	drone_sd	extract_physical
SHA-256:	beb7b3173644b5a8245f1ae715eda90da3639da617a4922b5afd35416f36fef8		
SHA-1:	faee1b8a3958c1d6386560eb56290656d0ad1f42		

2017_06_29_13_03_48.mp4	Videos	controller_ios	extract_logical
SHA-256:	23712c41078e8681e151b364d5103a318dc6826c32af23975e54f76dcae7dd39		
SHA-1:	4211ec27625959429df67993167e65bcd90d134		

2017_06_29_13_08_15.mp4	Videos	controller_ios	extract_logical
SHA-256:	9a1cac8fbd8a8a795a09c62e49be48f89b6f0472e1c4a6535b4e617d1ddf43644		
SHA-1:	fa562d027147633b9a3fe2c70bb39e6ad4a0ff8e		

2017_06_29_13_10_27.mp4	Videos	controller_ios	extract_logical
SHA-256:	5687d05405f0e3190646b65a3a4394a18e54d6a0154e0386324d84d77ce3daca		
SHA-1:	2d1cda5677ad571904d205f0accecedfc8854626		

2017_06_29_13_10_48.mp4	Videos	controller_ios	extract_logical
SHA-256:	47e11843d71a2b2c93e3876bc027ac29f60f3832cbc750dbbcbcb003b9101938		
SHA-1:	a43f967a28b060df4d3adb0fd8650515bee184af		

2017_06_29_13_11_55.mp4	Videos	controller_ios	extract_logical
SHA-256:	67c93467fd2c2aacc984faa8195adb7db766a04cc11d55f85174f580c018464		
SHA-1:	8e2a09cc3ca40362e8525bf399a4547a0eef8b67		

2017_06_29_13_13_04.mp4	Videos	controller_ios	extract_logical
SHA-256:	251b27c70c2ee73542d84208c461af8db1f6c05c02e75e87489008e30e0994be		
SHA-1:	e4ed30a4a854cd1fccec6fcbef4889c3ed285e7ea		

2017_06_29_13_13_49.mp4	Videos	controller_ios	extract_logical
SHA-256:	c374fa3fbfe9950c20c5fc19548ee831dbc15f1a4a01554acd71d25210204174		
SHA-1:	506f2676c9a185b88bd00362e32f6bcbb4a75d74		

DJI_0002.MOV	Videos	drone_sd	extract_physical
SHA-256:	8c0130901c10db776bb66aba8e3419a628d21c62bb208fbd9d02663ee668d6		
SHA-1:	3f043b78e31e45a4c0ea5687368c0f54741710bf		

DJI_0002_1.MOV	Videos	drone_sd	extract_physical
SHA-256:	b7e321ef952c4d417123970a2208663fc71b72b0b2c52068bd5bbc8cf3a511d8		
SHA-1:	4c6cc6c4e96836f07cce15495749cd62e69502bf		

DJI_0003.MOV	Videos	drone_sd	extract_physical
SHA-256:	5833218757f0835eb44ffa533672ec0259301d097d018a6502befc59bee2f3		
SHA-1:	965cabb36ce1a935353074cd7e8937ab71dc0233		

DJI_0003_1.MOV	Videos	drone_sd	extract_physical
SHA-256:	df988a711915fffb87a43de6877693c062dbf40d2549782d7291939fb2ad44d48		
SHA-1:	3b64c1648ddc6454e2b295bf94dbd09ea4431795		

DJI_0005.MOV		Videos	drone_sd	extract_physical
SHA-256:	e1d425f6213ecc84f6a68c13a8261d21acfd4d29bda3f74619d77973de0f7c ff			
SHA-1:	c6c2b79d523e66011fa2314a8ae9c01435bcf379			

DJI_0005_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	5ea89532ab3469120d74d3032d2564e63ad8b6ee42b02ce221c37d7072b256e6			
SHA-1:	60cd27bc08a8d9422b62fd032ba90fad8feb8371			

DJI_0006.MOV		Videos	drone_sd	extract_physical
SHA-256:	96b4e551824ffaf16a34254812778a25da71b41d93af663fe14647c64d0fc656			
SHA-1:	074aeaa3b24bc092aa482529eea96ab9010b4b6f			

DJI_0006_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	a796e8517cf21920e4667a415ee359f14da3f7f07a5fdda74d45b5f76c738514			
SHA-1:	39d075c2bac77ba15ed5c2d919033066d33d3dd3			

DJI_0009.MOV		Videos	drone_sd	extract_physical
SHA-256:	564e6bd58d751430e1a495a44803838717aebd0402e1da991043387d97207043			
SHA-1:	3fd13355ef8de7d3de68dcfc9933bd5f5e1b1009			

DJI_0009_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	b394e548cdf1e89eb132beb1f1559f69de9b2e6799fd6ad7fc70d571c36c2bff			
SHA-1:	02c0a75b2755e5dc7e39e505b9b6f5a36656911d			

DJI_0011.MOV		Videos	drone_sd	extract_physical
SHA-256:	2df4a3a5569fb636b7b982eed5386d52221e18c14cbd36b6dca1a70f8fbdeb2f			
SHA-1:	cec990bad5160cfabf5f32b139d2182da0d80a24			

DJI_0011_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	2a11dd32eb3b2da78f99ce368820648d54b3512d61de05f5bbc1c917546f7bee			
SHA-1:	c8d7b3ea1b3fc203da815aa6186a48991ff307bd			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Ios

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	2	2	0
Databases	12	0	0
Flight Logs	3	0	0
Flight Records	1	1	1
Images	7	7	7
Videos	7	7	0

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	17	17	0
Videos	12	12	0

Drone Flight Storage

Category	P3 Extracted	P4 Decoded	P5 Normalised
Drone Logs	11	22	11

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-19 11:15:06 mmls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: mmls.exe df006_external.001

2026-06-19 11:15:06 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p -o 8192 df006_external.001

2026-06-19 11:15:06 mmls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: mmls.exe df006_internal.001

2026-06-19 11:15:14 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p -o 63 df006_internal.001

2026-06-19 11:16:33 parser_ios v- rc=n/a

cmd: DF006_flight.zip C:\Users\Florin\Documents\dfdof_output\ft-df006-2017-ios-2\p2_image_parsing\controller_ios_parsed

out: controller_ios_parsed

2026-06-19 11:16:34 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796677

out: DJI_0001.JPG

2026-06-19 11:16:34 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796678

out: DJI_0002.MOV

2026-06-19 11:16:47 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796679

out: DJI_0003.MOV

2026-06-19 11:17:00 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796680

out: DJI_0004.JPG

2026-06-19 11:17:08 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796681

out: DJI_0005.MOV

2026-06-19 11:17:30 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796682

out: DJI_0006.MOV

2026-06-19 11:17:42 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796683

out: DJI_0007.JPG

2026-06-19 11:17:42 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796684

out: DJI_0008.JPG

2026-06-19 11:17:44 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796685

out: DJI_0009.MOV

2026-06-19 11:17:46 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796686

out: DJI_0010.JPG

2026-06-19 11:17:54 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796687

out: DJI_0011.MOV

2026-06-19 11:18:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796688

out: DJI_0012.JPG

2026-06-19 11:18:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796689

out: DJI_0013.JPG

2026-06-19 11:18:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796690

out: DJI_0014.JPG

2026-06-19 11:18:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796691

out: DJI_0015.JPG

2026-06-19 11:18:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796692

out: DJI_0016.JPG

2026-06-19 11:18:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796693

out: DJI_0017.JPG

2026-06-19 11:18:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796694

out: DJI_0018.JPG

2026-06-19 11:18:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796695

out: DJI_0019.JPG

2026-06-19 11:18:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796696

out: DJI_0020.JPG

2026-06-19 11:18:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796697

out: DJI_0021.JPG

2026-06-19 11:18:03 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796698

out: DJI_0022.JPG

2026-06-19 11:18:03 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 796699

out: DJI_0023.JPG

2026-06-19 11:18:03 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 1044485

out: DJI_0002_1.MOV

2026-06-19 11:18:04 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 1044486

out: DJI_0003_1.MOV

2026-06-19 11:18:06 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 1044487

out: DJI_0005_1.MOV

2026-06-19 11:18:08 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 1044488

out: DJI_0006_1.MOV

2026-06-19 11:18:09 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 1044489

out: DJI_0009_1.MOV

2026-06-19 11:18:10 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df006_external.001 1044490

out: DJI_0011_1.MOV

2026-06-19 11:18:11 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 5

out: FLY000.DAT

2026-06-19 11:18:11 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 6

out: FLY001.DAT

2026-06-19 11:18:12 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 8

out: FLY002.DAT

2026-06-19 11:18:12 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 9

out: FLY003.DAT

2026-06-19 11:18:12 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 10

out: FLY004.DAT

2026-06-19 11:18:12 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 11

out: FLY005.DAT

2026-06-19 11:18:13 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 12

out: FLY006.DAT

2026-06-19 11:18:13 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 13

out: FLY007.DAT

2026-06-19 11:18:14 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 14

out: FLY008.DAT

2026-06-19 11:18:14 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 15

out: FLY009.DAT

2026-06-19 11:18:14 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 63 df006_internal.001 16

out: FLY010.DAT

2026-06-19 11:18:16 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTLogToCSVtool.exe DJIFlightRecord_2017-06-29__13-00-45_.txt DJIFlightRecord_2017-06-29__13-00-45_.csv

out: DJIFlightRecord_2017-06-29__13-00-45_.csv

2026-06-19 11:18:17 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_13_03_48.thumbnail

out: 2017_06_29_13_03_48_exif.json

2026-06-19 11:18:17 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_13_08_15.thumbnail

out: 2017_06_29_13_08_15_exif.json

2026-06-19 11:18:18 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_13_10_27.thumbnail
out: 2017_06_29_13_10_27_exif.json

2026-06-19 11:18:18 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_13_10_48.thumbnail
out: 2017_06_29_13_10_48_exif.json

2026-06-19 11:18:18 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_13_11_55.thumbnail
out: 2017_06_29_13_11_55_exif.json

2026-06-19 11:18:19 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_13_13_04.thumbnail
out: 2017_06_29_13_13_04_exif.json

2026-06-19 11:18:19 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_13_13_49.thumbnail
out: 2017_06_29_13_13_49_exif.json

2026-06-19 11:18:20 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_13_03_48.mp4
out: 2017_06_29_13_03_48_exif.json

2026-06-19 11:18:20 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_13_08_15.mp4
out: 2017_06_29_13_08_15_exif.json

2026-06-19 11:18:20 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_13_10_27.mp4
out: 2017_06_29_13_10_27_exif.json

2026-06-19 11:18:21 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_13_10_48.mp4
out: 2017_06_29_13_10_48_exif.json

2026-06-19 11:18:21 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_13_11_55.mp4
out: 2017_06_29_13_11_55_exif.json

2026-06-19 11:18:22 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_13_13_04.mp4
out: 2017_06_29_13_13_04_exif.json

2026-06-19 11:18:22 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2017_06_29_13_13_49.mp4
out: 2017_06_29_13_13_49_exif.json

2026-06-19 11:18:22 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.JPG
out: DJI_0001_exif.json

2026-06-19 11:18:23 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.JPG
out: DJI_0004_exif.json

2026-06-19 11:18:23 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0007.JPG
out: DJI_0007_exif.json

2026-06-19 11:18:23 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0008.JPG

out: DJI_0008_exif.json

2026-06-19 11:18:23 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0010.JPG

out: DJI_0010_exif.json

2026-06-19 11:18:24 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0012.JPG

out: DJI_0012_exif.json

2026-06-19 11:18:24 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0013.JPG

out: DJI_0013_exif.json

2026-06-19 11:18:24 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0014.JPG

out: DJI_0014_exif.json

2026-06-19 11:18:24 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0015.JPG

out: DJI_0015_exif.json

2026-06-19 11:18:25 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0016.JPG

out: DJI_0016_exif.json

2026-06-19 11:18:25 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0017.JPG

out: DJI_0017_exif.json

2026-06-19 11:18:25 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0018.JPG

out: DJI_0018_exif.json

2026-06-19 11:18:26 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0019.JPG

out: DJI_0019_exif.json

2026-06-19 11:18:26 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0020.JPG

out: DJI_0020_exif.json

2026-06-19 11:18:26 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0021.JPG

out: DJI_0021_exif.json

2026-06-19 11:18:27 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0022.JPG

out: DJI_0022_exif.json

2026-06-19 11:18:27 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0023.JPG

out: DJI_0023_exif.json

2026-06-19 11:18:27 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.MOV

out: DJI_0002_exif.json

2026-06-19 11:18:28 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002_1.MOV

out: DJI_0002_1_exif.json

2026-06-19 11:18:28 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.MOV

out: DJI_0003_exif.json

2026-06-19 11:18:29 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003_1.MOV
out: DJI_0003_1_exif.json

2026-06-19 11:18:29 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.MOV
out: DJI_0005_exif.json

2026-06-19 11:18:30 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005_1.MOV
out: DJI_0005_1_exif.json

2026-06-19 11:18:30 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0006.MOV
out: DJI_0006_exif.json

2026-06-19 11:18:31 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0006_1.MOV
out: DJI_0006_1_exif.json

2026-06-19 11:18:31 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0009.MOV
out: DJI_0009_exif.json

2026-06-19 11:18:32 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0009_1.MOV
out: DJI_0009_1_exif.json

2026-06-19 11:18:32 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0011.MOV
out: DJI_0011_exif.json

2026-06-19 11:18:32 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0011_1.MOV
out: DJI_0011_1_exif.json

2026-06-19 11:18:51 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY000.DAT
out: FLY000.csv, FLY000.kml

2026-06-19 11:19:08 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY001.DAT
out: FLY001.csv, FLY001.kml

2026-06-19 11:19:24 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY002.DAT
out: FLY002.csv, FLY002.kml

2026-06-19 11:19:42 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY003.DAT
out: FLY003.csv, FLY003.kml

2026-06-19 11:20:00 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY004.DAT
out: FLY004.csv, FLY004.kml

2026-06-19 11:20:18 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY005.DAT
out: FLY005.csv, FLY005.kml

2026-06-19 11:20:36 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY006.DAT

out: FLY006.csv, FLY006.kml

2026-06-19 11:20:54 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY007.DAT

out: FLY007.csv, FLY007.kml

2026-06-19 11:21:14 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY008.DAT

out: FLY008.csv, FLY008.kml

2026-06-19 11:21:32 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY009.DAT

out: FLY009.csv, FLY009.kml

2026-06-19 11:21:49 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe FLY010.DAT

out: FLY010.csv, FLY010.kml

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap ≥ 60 s AND $\geq 90\%$ of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs ≤ 25 m with ≥ 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a ± 2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.